# A Quantitative Text Analysis of Addiction Framing by United States Governors, 1999 - 2024

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## INTRODUCTION

Drug overdose death rates in the United States have steadily increased between 1999 and 2022, peaking in 2022 before beginning to decline slightly (Kaiser Family Foundation, 2025). Driven in large part by the introduction and subsequent overprescribing of opioid painkillers, addiction and related deaths have reached epidemic levels across the country (Singh *et al.*, 2019). Despite growing research suggesting that addiction bears resemblance to a disease, and is not, in fact, a result of any individual, moral failing, there remains significant stigma associated with people who use drugs or have substance use disorder (SUD) (Detar, 2011; Berridge, 2017; Matthews, Dwyer and Snoek, 2017; Baker *et al.*, 2020). Americans tend to hold more negative attitudes toward people with SUD compared to people with mental illness and also demonstrate lower support for policies aimed at supporting those with SUD versus policies aimed at addressing mental illness (Barry *et al.*, 2014). The mainstream language and rhetorical framing of issues related to SUD have changed over time to try to better reflect the fact that addiction is a pathology in order to reduce stigma (NIDA). However, stigmatized language around SUD still exists and influences policy decisions, leading to addiction services and treatments being both underresearched and underresourced (Richter, Vuolo and Salmassi, 2019). Understanding the different ways in which SUD and related treatments and policies are framed can provide insights into what factors affect framing. Thus, the research question that this paper will evaluate is:

***Do partisanship and public health landscapes shape how states frame addiction and addiction treatment services?***

Public attitudes toward and policy framing of addiction and related services have shifted dramatically over the past several decades (Provine, 2011). The war on drugs beginning during President Nixon's administration in the 1970s solidified a criminalized framing of substance use, couching substance use in highly racialized and discriminatory terms (Provine, 2011; Sirin, 2011). In addition to framing substance use in racist terms, criminalized framing of addiction emphasized the need for strict policing, penalties, and sentencing for both people selling and using drugs: "drug addicts" were painted as dangerous criminals without moral integrity (Provine, 2011, pp. 43). Federal policies were shaped by and reflective of this framing toward addiction: the institution of mandatory minimums for possession of crack cocaine in 1986 represented one of the most stark instances of the racialized criminalization of substance use and related activities (Provine, 2011, pp. 45-46). This criminalized framing of drugs and people who use drugs has harmful consequences in terms of increasing racially-motivated and stigma-motivated discrimination. In addition to the very real impacts of discrimination, criminalized framing also leads to policymaking that fails to address the root causes of SUD and ultimately does not support evidence-based recovery and treatment services (Wild *et al.*, 2017). Criminalized framing often places blame on individuals with SUD and emphasizes policing and incarceration as solutions to SUD.

Many of the consequences and stereotypes of the war on drugs approach to the criminalization of substance use persist within contemporary policy landscapes and rhetoric (Provine, 2011). Still, the public health landscape with regards to addiction and SUD treatments is rapidly evolving both in the U.S. and globally (Hawk, Vaca and D'Onofrio, 2015; Wild et al., 2017). Recognizing the societal, environmental, and genetic factors that contribute to SUD, there have been strides toward recognizing SUD as a public health crisis rather than a strictly criminal issue (Nadelmann and LaSalle, 2017). Emerging in part from the public health perspective on SUD are harm reduction strategies, which attempt to reduce adverse effects of substance use, such as overdose, contraction of blood-borne

disease, or death, rather than approach substance use from an "abstinence-only" perspective (Single, no date). Harm reduction strategies contain a wide range of services, including clean needle distribution, medication-assisted treatment for maintenance or recovery, supervised consumption, and access to overdose-reversal drugs such as naloxone (Wild *et al.*, 2017). While certainly not comprehensive, the federal government has taken some steps toward a public health-guided harm reduction approach to SUD beginning with the Obama administration (Nadelmann and LaSalle, 2017). State implementation of harm reduction strategies is extremely varied, with states such as Vermont taking a much more public health approach in utilizing multiple harm reduction strategies and states such as Texas still relying on a criminalized approach to substance use (CDC, 2026; The Network for Public Health Law, 2024). Despite evidence that harm reduction strategies can save lives and better support people with SUD, there remains substantial stigma associated with harm reduction policies, which translate to ideological opposition and a lack of political will to implement these types of policies (Childs *et al.*, 2021). In comparison to a criminalized framing of substance use, a public health approach emphasizes the external factors that lead to SUD and also the potential for recovery through evidence-backed public health interventions.

As with almost all policy issue areas, there is a clear partisan divide in both healthcare and public health perspectives (Clinton et al., 2020; Levitt, 2016). Despite significantly different partisan landscapes and populations, the vast majority of U.S. states experienced an increase in drug-related death rates, and particularly opioid-related death rates, during the period from 1999 to 2024 (Kaiser Family Foundation). Governors' State of the State (SOTS) addresses, usually delivered annually, provide a useful measure both of key issues affecting a given state as well as the framing of those issues within the state context (Picard and Stammbach, 2022). Considering theories of values-based policymaking, the drug- and alcohol-related death rate as a proxy for the state's public health landscape may affect how state leaders frame addiction and related issues (Shams *et al.*, 2021). Using the public health versus criminalization theoretical framework laid out by Riboni (2019), this paper will examine 1) whether governors are talking about addiction and 2) when they are, their framing of the issue. Understanding whether factors such as party affiliation and the state's drug-related death rate affect the framing of addiction could offer insight into how to most effectively communicate about these topics in order to work toward reducing stigma and ultimately saving lives. Therefore, the hypotheses for this research are:

***H1: Partisanship affects whether state leaders frame addiction & treatment from a public health perspective or a criminalized perspective.***

***H2: Increasing drug-related death rates correspond to a shift in state leaders' framing of addiction from a criminalized perspective toward a public health perspective.***

***H3: The framing of addiction and related services by state leaders shifts from criminalized to public health over time.***

**DATA AND METHODOLOGY**

In order to evaluate the above hypotheses, the author built upon a dataset of U.S. SOTS speeches originally collected by Léo Picard and Dominik Stammbach (Picard and Stammbach, 2022). As noted by Picard and Stammbach (2022), SOTS addresses are useful units of analysis to understand state legislative priorities as they are the primary public addresses that are directed toward the entire state electorate. The corpus used for analysis consisted of 1252 SOTS and State of the Union (SOTU)

speeches from 1999-2024. Some states do not have a SOTU every year, so in some cases either state of the budget or inaugural speeches are included instead. The speeches represent 237 unique governors (or presidents in the case of SOTUs), with 549 speeches given by democratic governors (43.8%) or presidents and 690 (55.1%) speeches given by republican governors or presidents. Thirteen speeches (1.1%) were given by a governor of another party (labelled "Other" in figures and tables). After preprocessing the 1252 speeches, including removing English stopwords, punctuation, and compounding key terms of interest for analysis, the text corpus consisted of 19,720 unique features. Supplementary data include the age-adjusted drug- and alcohol-related death rates per state (CDC Wonder). Dataset and variable descriptions can be found in Table A1 in the appendix.

Three different methods were applied to analyze the corpus and evaluate the hypotheses. First, keyword-assisted topic modeling was conducted to understand the major topics discussed by governors and to observe the partisan and over-time changes in these topics. Keyword assisted topic modeling is useful to identify themes within a large corpus, and keywords make interpretability more clear (Eshima, Imai and Sasaki, 2024). Next, the relevant addiction- and treatment-related terms were identified within the corpus using dictionary filtering, and the corpus was subsetted to include only speeches that featured one or more of these topical terms (see full keyword list, based on existing literature, in the appendix). After subsetting this corpus to identify relevant speeches, there were 584 speeches, with every state and year still represented within the subset. The resulting subset was used to calculate framing scores using Latent Semantic Scaling (LSS). LSS is a semisupervised document scaling method which relies on researcher-provided seed words to create a continuous, bipolar scale (Watanabe, 2021). Deriving seed words from the literature, LSS was used to produce document framing scores for each speech, with "criminalized" on the negative pole and "public health" on the positive pole (Table A4 in the Appendix lists seed words for each frame). Finally, a regression analysis with document LSS scores as the dependent variable was conducted to evaluate the hypotheses.

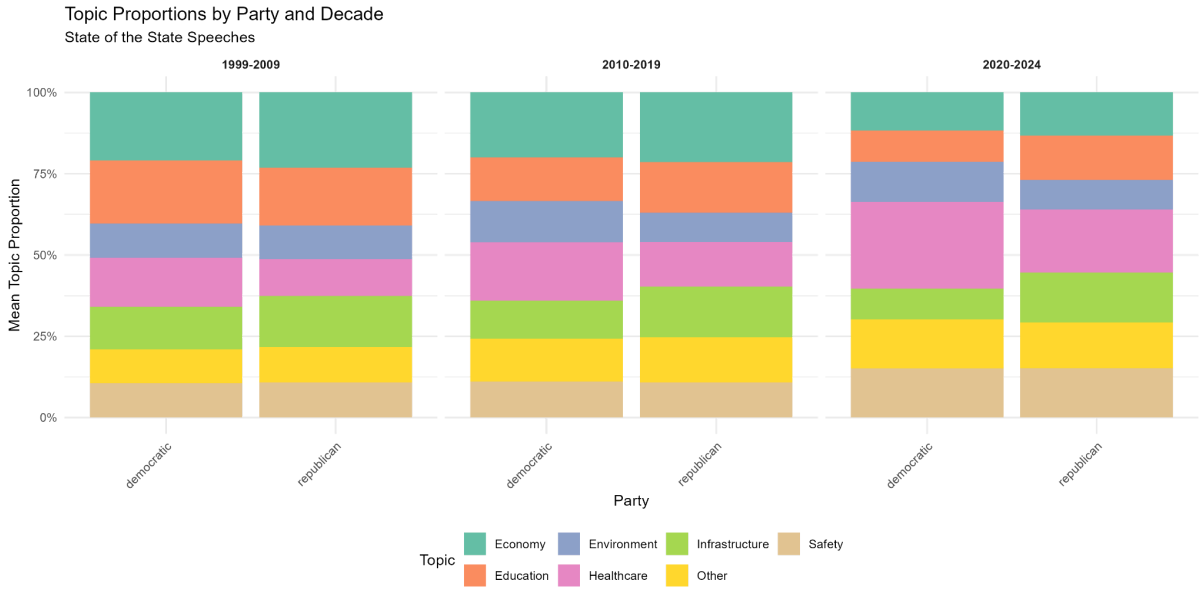
In order to validate the scores produced through LSS, a random sample of 400 speech chunks was annotated by the author based on a predetermined codebook, which can be found in the appendix. Sentence chunks that were irrelevant to the topic were excluded from validation analysis. Hand-coded chunks were scored along an ordinal scale while LSS scores are continuous, so Spearman's correlation was calculated between human-annotated chunks and LSS scored chunks. Spearman's correlation coefficient is useful when there is no assumption of a linear relationship and when there is an ordinal variable involved (Hauke and Kossowski, 2011). For this validation, Spearman's rho was calculated as 0.55 with a p-value of 0.0004616, representing a moderate positive correlation between hand-coded and LSS scores. See Figure A1 in the appendix for a plot of human-coded scores versus LSS scores. All data wrangling, analysis, and visualization was done in R, and code file names and descriptions can be found in Table A2 in the appendix.

## RESULTS

### Topic Modeling and Subsetting

Topic modeling revealed clear trends in the main topics discussed by governors in their addresses. Figure 1 depicts the topic modeling results in average topic proportions for republican and democratic governors over three decades. Initial topic modeling was attempted utilizing addiction-relevant terms as keywords, but it was quickly apparent that addiction and related topics were discussed too infrequently across the entire corpus to form a single, coherent category.

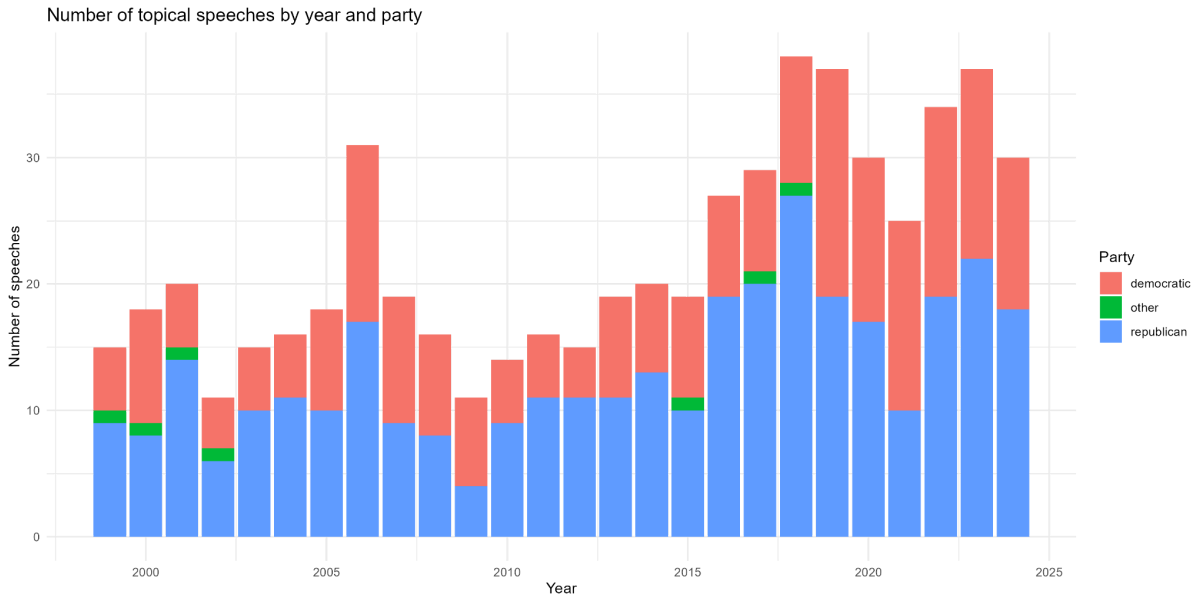
**Figure 1. Full corpus topic proportions**




Unsurprisingly, economic topics remain prevalent in both parties' speeches over time. Education seems to be discussed less by both parties over time, while healthcare is discussed more over time and relatively more by democrats compared to republicans. While the 2020-2024 period represents fewer data than the preceding decades, it seems that democrats are focusing much more on healthcare. Overall, there are not significant differences among topics discussed by party, likely because these topics represent key political issues across all states.

After filtering the corpus of speeches for only speeches that reference one or more relevant keywords, the corpus subset of 584 speeches was divided by party roughly proportionally to the split of the full corpus.

**Figure 2. Topical speeches by party**



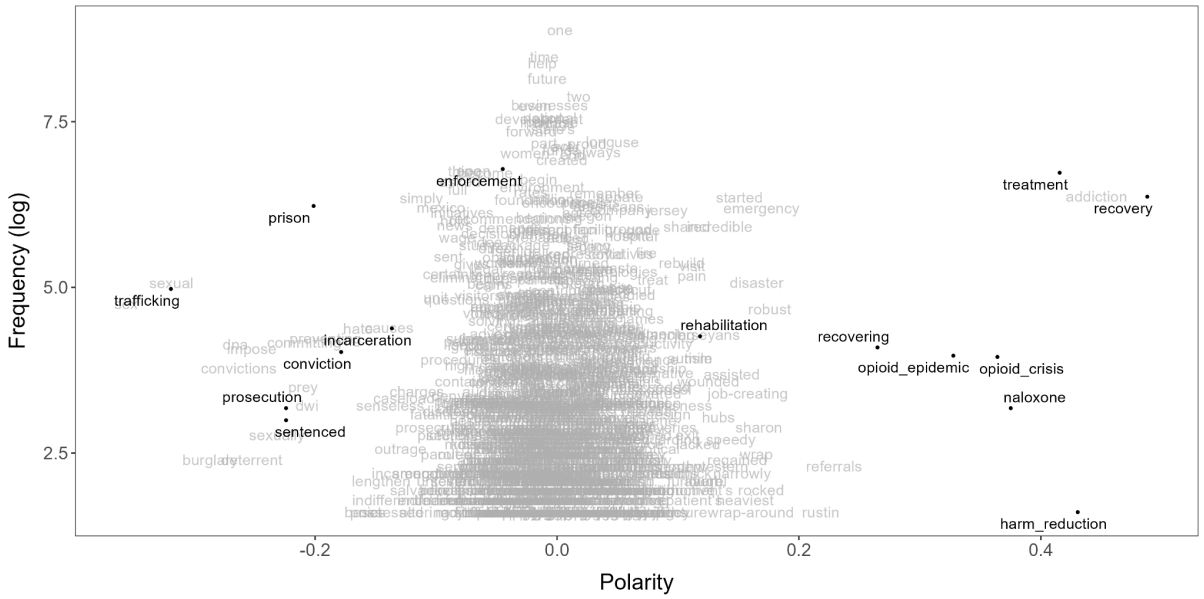

Between 1999 and 2024, governors included more references to addiction, treatment, and related issues in their SOTS addresses. From 2010 to 2019, democratic governors dominated the number of

topical speeches, while the divide becomes more even between parties from 2020 to 2024. The year during which the most speeches featured addiction or related terms was 2018, which might be explained by a sudden jump in overdose death rates, from 16.3 people per 100,000 in 2015 to 21.7 people per 100,000 in 2017 (Kaiser Family Foundation). Despite the relatively equal partisan split between these topical speeches, it is worth noting that less than half (46.6%) of all SOTS addresses from 1999 - 2024 mention addiction or related terms, despite increasing drug- and alcohol-related death rates during the same period (Kaiser Family Foundation).

**Latent Semantic Scaling**

While the relative infrequency with which state leaders discuss topics related to addiction is interesting, the framing used to discuss these topics has greater implications for future policymaking on this issue. In order to evaluate the framing of these speeches, an LSS model was fitted to the subsetted corpus to produce document scores for each speech. To measure public health framing, seed words such as "recovery", "opioid_epidemic", and "harm reduction" were supplied to the model; for the criminalized frame, seed words such as "enforcement", "crackdown", and "sentenced" were supplied. Figure 3 represents the polarity of terms within the corpus subset, with the x-axis representing polarity scores along the public health-criminalized scale and the y-axis representing logged term frequency, with seed words in bold.

**Figure 3. Polarity scores vs term frequency**




Terms associated with public health framing are scored positively while terms associated with criminalized framing are scored negatively. The vast majority of terms are clustered around zero, which suggests that most of the terms in the corpus are neutral along these dimensions. This makes sense since SOTS addresses often cover a wide range of topics, and addiction-related topics might only make up a very small section of a given address. The distribution is slightly right skewed, suggesting that there may be slightly more public health framing terms relative to criminalized terms across the entire corpus subset. Table A4 in the appendix depicts a list of terms with the highest/lowest polarity scores for each pole as scored by the LSS model.

To visualize how framing has changed over time and evaluate framing differences by party, LSS speech scores were averaged across all democratic governors and all republican governors per year. Figure 4 depicts these trends with year on the x-axis and average LSS score on the y-axis, with colored ribbons representing standard errors.

**Figure 4. Average LSS scores by party over time**

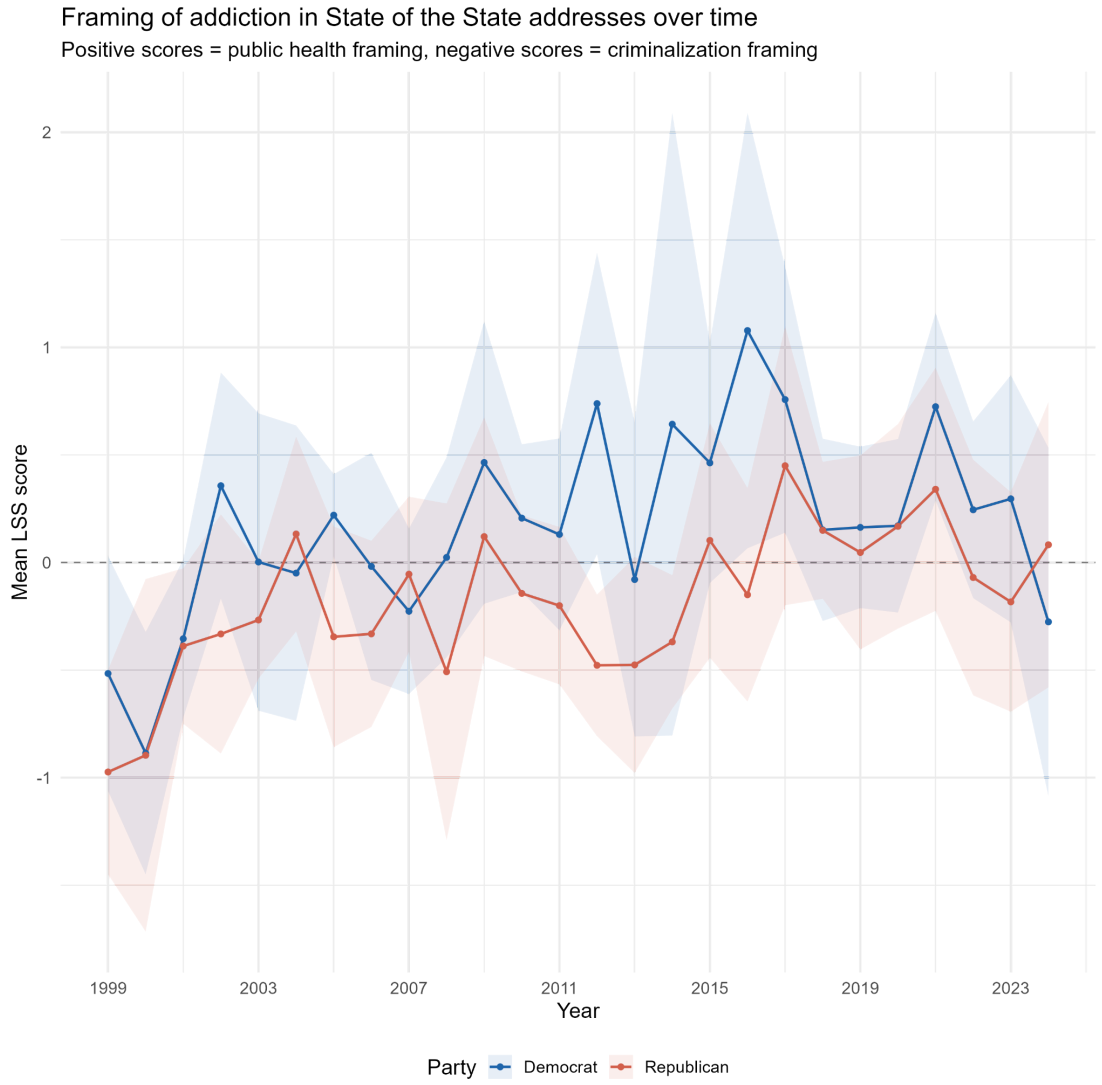



Figure 4 demonstrates that, for the period 1999 to 2024, the average LSS score for democratic governors was almost always positive and greater than the average LSS score for republican governors. This indicates that, on average, democratic governors tend to use a public health frame to discuss addiction and related issues, while republic governors tend to use a criminalized frame to discuss the same topics. Interestingly, the average framing flips in 2024: democratic governors utilized more criminalized framing while republicans used slightly public health-focused framing. Additionally, both democrats' and republicans' average LSS scores trend slightly upward over time; while the trend is far from linear, this represents a slight shift toward public health framing over time.

**Regression Analysis**

Three regression models were fitted to evaluate the hypotheses. First, following Primo et al. (2007), an Ordinary Least Squares (OLS) model with clustered standard errors was fitted to estimate the effect of partisanship on LSS framing scores. Next, based on Millimet and Bellemare (2023), an OLS model

was fitted to take into account within-state fixed effects to estimate the effect of states' drug- and alcohol-related death rates on issue framing. Finally, the full model estimates effects of both partisanship and drug- and alcohol-related death rate with standard errors clustered by state. All models hold year constant.

**Table 1. Regression results**

| | H1: Party | H2/H3: Within-State | Full Model |
|---|---|---|---|
| (Intercept) | −54.537*** | | −47.037 |
| | (13.535) | | (28.782) |
| Other party | 0.080 | | 0.080 |
| | (0.130) | | (0.130) |
| Republican | −0.309** | | −0.302** |
| | (0.110) | | (0.111) |
| Year | 0.027*** | 0.011 | 0.023 |
| | (0.007) | (0.010) | (0.014) |
| Drug-related death rate | | 0.008 | 0.003 |
| | | (0.005) | (0.008) |
| Num.Obs. | 569 | 569 | 569 |
| R2 | 0.064 | 0.481 | 0.065 |
| R2 Within | | 0.057 | |

* p < 0.05, ** p < 0.01, *** p < 0.001
Standard errors clustered by state.

Model 1 (H1: Party) reveals that there is a statistically significant relationship between party and framing addiction along a public health-criminalized scale. Republican state leaders scored 0.309 points lower than the reference group, democratic governors, on the LSS scale, indicating more criminalized framing than democratic governors. Model 2 (H2/H3: Within-State) does not yield any statistically significant findings. The $R^2$ Within of 0.06 indicates that, after accounting for state differences, partisanship and the drug-related death rate explain only 6% of variation in LSS scores. Finally, Model 3 (Full Model) finds that when accounting for within-state correlation in errors, partisanship still has a statistically significant relationship to the framing of addiction and related policies. This is indicated by the negative and statistically significant coefficient (-0.302) for republican state leaders. While the coefficient for year is statistically significant in Model 1, there is no significant relationship between year and framing when accounting for state fixed effects and clustered variance in Models 2 and 3. Also contrary to theoretical expectations, the drug-related death rate does not have a statistically significant effect on framing of addiction. As a robustness check, Models 2 and 3 were re-run with the age-adjusted death rate lagged by one year to see whether the

death rate had a delayed effect on framing. The lagged death rate still did not have a statistically significant effect on framing - Table A6 in the appendix reports these regression results. Overall, the null hypothesis for H1 can be rejected, while the null hypotheses for both H2 and H3 cannot be rejected. See Figure A2 in the appendix for a plot of coefficients from Models 1 and 2.

## CONCLUSION

In conclusion, this research has demonstrated that when state leaders discuss addiction and related policies, the framing that they use depends on their partisanship. After identifying governors' SOTS addresses which contained terms relevant to addiction, treatment services, or penalties, speeches were scored using LSS along a continuous scale of public health framing and criminalized framing. Speech scores reveal that there is a relationship between the party of the governor and whether the governor frames addiction and related policies from a public health perspective or from a criminalized perspective. While observing framing trends over time suggests that both parties might be moving closer to a public health framing of addiction, regression analysis resulted in mixed findings regarding the role of time. Contrary to theoretical expectations posited in H2, the public health landscape, as represented by drug- and alcohol-related deaths, does not have a statistically significant effect on how governors frame addiction. Overall, these results suggest that state leaders within the two major parties in the U.S. are not aligned on how to talk about addiction and services to support people with SUD. Differences among state leaders' framing of these issues will perpetuate a state landscape where harm reduction policies and resources are inequitably distributed and accessible. Like many divisive political issues, partisan polarization on this topic will likely lead to slow progress at the federal level.

Limitations of this research include the conflation of general crime-related terms with terms that apply specifically to addiction and related policies. While certain filtering keywords and seed words are only applicable to the context of addiction (for example, "opioid epidemic"), other terms like "sentenced" can apply to unrelated criminal activities. Additionally, due to the length of SOTS addresses and the relatively few references to addiction topics, the scaling scores are influenced by speech noise. Generally, as noted by Watanabe (2020), LSS also lacks the sensitivity of dictionary-based approaches to text analysis. Further research on this subject could apply a similar scale to texts focused only on addiction in order to more accurately assess the framing differences in a topic-specific context. Seed words and scaling should be validated by additional coders in order to test reliability and replicability. Additionally, an extension of this research could include case studies on specific states with particularly strong framing or distinct framing patterns. It is important to continue to study the framing of issues like addiction in a political context in order to understand what drives polarization and disagreement on these highly salient topics.

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**APPENDIX**

**Table A1.** Data

| File Name | Dataset | Source | Description | Variables |
|---|---|---|---|---|
| sots_data.csv | U.S. State of the State speech transcripts, 1995 - 2024 | https://github.com/leops95/sots/blob/master/README.md and data collected by author (2023 - 2024 speeches) | Includes full text of State of the State, Budget, and Inaugural addresses for all 50 states plus U.S. States of the Union | file_id = file ID corresponding to state and year of speech<br><br>state_id = 2-letter state abbreviation<br><br>state_name = full state name<br><br>speaker = governor or president delivering address<br><br>party = political party of speaker<br><br>type = type of speech, (sots = state of the state; budg = budget address, inaug = inaugural address)<br><br>quality = remarks either as prepared or as delivered<br><br>text = full speech text<br><br>source = source of text |
| od_deaths_1999_2024.csv | CDC Wonder Cause of Death data, 1999 - 2024 | https://wonder.cdc.gov/wonder/help/ucd.html# | Age-adjusted drug and alcohol-related deaths by state | year = year<br><br>state = state<br><br>population = population |

| | | | | |
|---|---|---|---|---|
| | | | | age_adj_rate = age-adjusted death rate |

**Table A2.** Code files and descriptions

| File Name | Description |
|---|---|
| sots_merge.r | Code used to merge Picard and Stammbach (2022) data with author-collected data - output file is sots_data.csv |
| od_data.r | Code used to clean and aggregate drug- and alcohol related death rate data from CDC WONDER - output file is od_deaths_1999_2024.csv |
| sots_topics.r | Code to preprocess corpus, run topic models, and create Figures 1 and 2 |
| sots_lss.r | Code to take validation sample for handcoding, run LSS models, and predict document-level polarity scores. Also includes code to create Figures 3 and A1. |
| sots_regression.r | Code to run regression models using LSS document scores. Includes code for producing Figure 4, Table 1, and Figure A2. |

**Table A3.** Key terms used to subset corpus for speeches that reference addiction and related topics

| |
|---|
| "addict*", "overdose*", "overdose_deaths", "opioid*", "opioid_epidemic", "opioid_crisis", "fentanyl", "naloxone", "substance*", "substance_use_disorder", "substance-abuse", "epidemic", "epidemics", "harm_reduction", "drug_courts", "trafficking", "traffickers", "heroin", "methamphetamine", "meth", "cocaine" |

**Table A4. Latent Semantic Scaling seed words**

| **Public health frame** | **Criminalized frame** |
|---|---|
| "treatment", "recovery", "rehabilitation", "naloxone", "harm_reduction", "opioid_epidemic", "opioid_crisis", "recovering" | "incarceration", "prison", "conviction", "sentenced", "enforcement", "crackdown", "prosecution", "trafficking" |

**Table A5. Top polarity words by frame**

| Public health frame | Polarity score | Criminalized frame | Polarity score |
|---|---|---|---|
| recovery | 0.4880037 | crimes | -0.3964027 |
| addiction | 0.4334070 | sentences | -0.3587013 |
| harm_reduction | 0.4305115 | sex | -0.3562894 |
| treatment | 0.4155612 | possession | -0.3561629 |
| disease | 0.4088035 | convicted | -0.3553388 |
| reinvented | 0.3970194 | criminals | -0.3532510 |
| opioid | 0.3918715 | felons | -0.3508944 |
| shame | 0.3756608 | felony | -0.3469906 |
| naloxone | 0.3751220 | sentence | -0.3392270 |
| narcan | 0.3675274 | gangs | -0.3362122 |
| opioid_crisis | 0.3640979 | trafficking | -0.3192123 |
| life-saving | 0.3532995 | sexual | -0.3188068 |
| battling | 0.3521587 | violent | -0.3143011 |
| doorway | 0.3486788 | firearm | -0.3135165 |

**Figure A1.** Hand-coding scores versus LSS scores

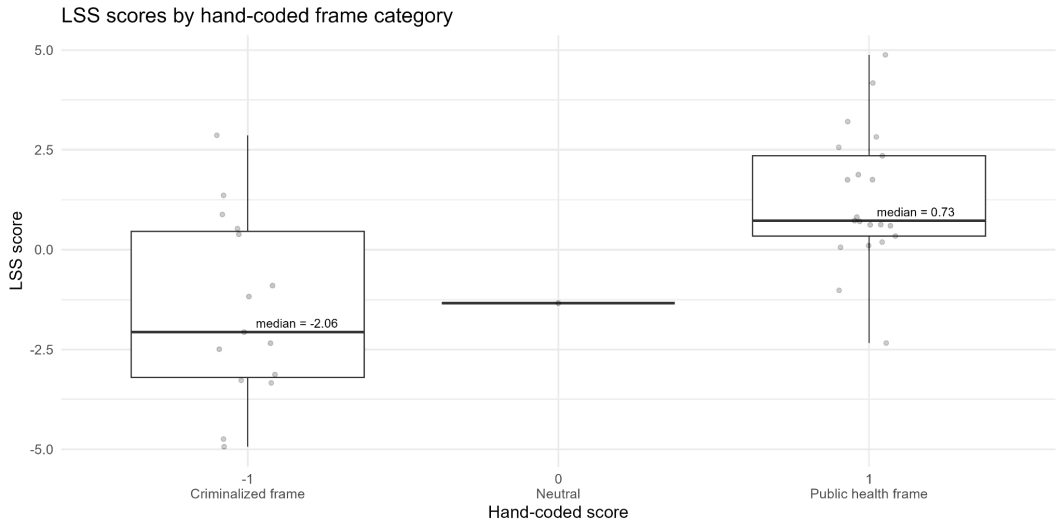


**Figure A2:** Regression coefficients plot

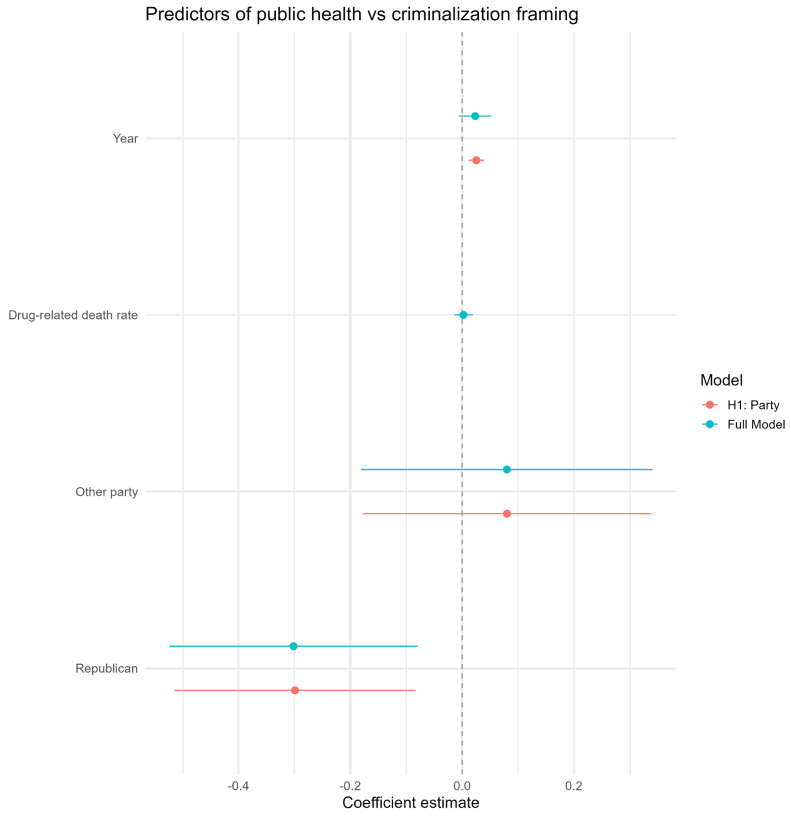



**Table A6.** Regression results with lagged death rate

| | H2/3 Lagged | Full Model Lagged |
|---|---|---|
| Lagged drug-related death rate | −0.003 | −0.001 |
| | (0.005) | (0.008) |
| Year | 0.026* | 0.024 |
| | (0.010) | (0.015) |
| (Intercept) | | −48.796 |
| | | (30.052) |
| Other party | | 0.163 |
| | | (0.131) |
| Republican | | −0.297* |
| | | (0.118) |
| Num.Obs. | 519 | 519 |
| R2 | 0.486 | 0.047 |
| R2 Within | 0.035 | |

* p < 0.05, ** p < 0.01, *** p < 0.001
Standard errors clustered by state.

**Codebook for human-coded validation sample**

*Unit of Analysis*

- Each coding unit is a chunk of 4 sentences from a U.S. State of the State Speech
- Code based on the dominant framing across the whole chunk
- If mixed, code the strongest or most emphasized framing

*Labels*

| **Label** | **Core Meaning** |
|---|---|
| 1: Public health frame | Addiction and related topics are considered a matter of public health rather than an individual or moral failing, and there is a focus on recovery and rehabilitation rather than punitive actions like incarceration |
| 0: Neutral | Chunks that don't have a dominant frame |
| -1: Criminalized frame | Addiction and related topics are considered an individual or moral failing, and there is an emphasis on punishing people participating in these activities rather than addresses external factors that may lead to substance use |
| 99: Off topic | Chunks that are not relevant to the topics at hand |

*Detailed coding rules and example patterns*

**1: Public health frame**

Core concepts: addiction or drugs are discussed in a way that focuses on addiction as a health issue more so than a crime.

- There will be less moral language, but if there is moral language, it will be more positive than negative
- There might be clinical or medical language
- The victim or person using substances is not attributed blame or shame
- There is recognition of a public health emergency relating to substance use, such as "opioid crisis" or "opioid epidemic"
- There is a rehabilitative approach focused on health and safety
- There is blame attributed to entities other than the person or people using drugs (i.e. holding pharmaceutical companies accountable)
- There will be mention of harm reduction strategies, including:
    - Naloxone and other medication-assisted treatment options
    - Rehabilitation
    - Treatment

Example patterns:

"...the new Family Strong program which has (been) made available for free to support families who are impacted by someone in their family having a substance use disorder.We've also got a fantastic and amazing team inside of Human Services which is led by Chris Jones, the behavioral health team led by Pam Sagness. They've been amazing. They're doing amazing work and they've been an amazing partner with Recovery Reinvented."

**0: Neutral**

Core concepts: The framing is balanced equally between public health and criminalization.

- Both frames are equally present in the chunk

Example patterns:
"Personally, the next time someone offers me a ride on a horse in Africa, I'll take a seat at the bar instead. For most of us, what defines our lives is how we learn from those mistakes and move forward. I believe that should apply to everyone, even men and women who commit a crime."

**-1: Criminalized frame**

Core concepts: addiction is discussed primarily as a crime, with people using substances being discussed as criminals

- There might be more moral language, attempting to negatively frame people who use drugs
- There is blame or shame attributed to people who use drugs
- There is a punitive approach focused on incarceration or police activity rather than a rehabilitative approach
- There are highly stigmatized words used to describe people who use drugs or related behavior

Example patterns:
"Our corrections institutions are filling with people who, losing a battle against addiction, resort to increasingly violent crimes."

"Law and order requires strong borders. The crisis at the US-Mexico border over the past year has witnessed staggering illegal migration and a massive influx of narcotics like fentanyl."

**99: Off topic**

Core concepts: The chunk does not mention any relevant topics.

- Any policy topic unrelated to public health, addiction, addiction treatment, drugs, drug laws, or crime.

Example patterns:
"And we need to do more to provide every student with the broad range of opportunities and experiences that are available in first-rate after-School programs. These programs give working parents the security of knowing that their children have a safe, educational after-school environment."

“We cannot continue our record economic trajectory if our logging trucks can't get across old bridges or we can't get our farm products to market. My budget invests another $200 million in ongoing funding to fully address our known maintenance needs locally and statewide.”

**Table A7**. AI Usage Appendix

| AI Tool | Explanation | Prompt Used |
|---|---|---|
| Claude | Workflow organization | “I'm interested in both "predicting" how much a governor talks about the relevant topics and also framing. Would this workflow make sense?<br>1. preprocess corpus<br>2. keyword frequency based on defined set of relevant terms<br>3. keyATM analysis using lagged drug/alcohol death rate & partisanship as covariates<br>4. dictionary-based sentiment analysis” |
| Claude | Keyword list clarity | I have these keywords identified for two main topics. Do these lists make sense for a starting point in a keyATM model?<br>Harm reduction / health / medical<br>“safe injection”, “harm reduction”, “harm minimization”, “methadone maintenance”, “opioid maintenance”, “supervised injection”, “reducing risk”, “reducing harm”, “substance use disorder”, “risk reduction”, “syringe exchange”, “needle exchange”, “clean needle”, “substance misuse”, “opioid use disorder”, “opioid crisis”, “overdose crisis”, “supervised injection”, “needle distribution”, “syringe distribution”, “medication-assisted treatment”, “safer inhalation”, “low-threshold opioid substitution”, “syringe-exchange”, “overdose prevention”, “supervised |

| | | |
|---|---|---|
| | | consumption", "opioid treatment", "behavioral health", "opioid agonist therapy", "alcohol use disorder", "substance use recovery", "public-health emergency"<br><br>Abstinence-only / criminal / legal<br>"war on drugs", "addict", "drug abuser", "drug war", "drug crime", "drug dealer", "junkie", "substance abuser", "drug addict", "drug habit", "pill abuse", "prescription pill abuse", "alcoholic", "alcohol abuse", "former addict", "detox", "abstinence-only", "twelve step", "drug criminalization", "drug-related crime", "prescription drug abuse", "opioid abuse", "drug arrest", "unprescribed pill", |
| Claude | Validation | "What is the best way to validate my keyATM output as a single coder?" |
| Claude | Validation | "What is the best way to validate my latent semantic scaling output as a single coder?" |
| Claude | Regex help | "These words are included in my custom stopwords list that I use with tokens remove during tokenisation, but they are appearing in the top words of one of my topics from my topic model. "it's", "we've", "we're", "that's", "let's", "i'm", and "don't". What might be causing this?" |
| Claude | LSS workflow | "I have a random sample of text chunks that I hand-coded and have them labeled by chunk_id. In order to test my hand-coding scores against LSS scores, should I run an LSS model on my full corpus and then subset the code chunks that I randomly |

| | | |
|---|---|---|
| | | sampled?” |
| Claude | Statistical models | “These are my hypotheses. What types of statistical models make sense to run regressions?<br>H1: Partisanship affects whether state leaders frame addiction & treatment from a public perspective or a criminal/punitive perspective.<br><br>H2: Increasing drug-related death rates corresponds to a shift in the framing of addiction from a criminal/punitive perspective toward public health.<br><br>H3: Framing shifts from criminal/punitive to public health over time.” |
| Claude | Statistical robustness check | “I ran regression models with a variable normally and with the variable lagged by one year and neither yielded statistically significant results. Should I report the results of both models even though the results didn’t change in terms of significance?” |